

Predictive Zonotope Reduction: Deep-Research Brief

Project notes synthesized from `science/SCIENCE.md` and `results/tacas-main`

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Abstract

This brief frames Predictive Zonotope Reduction as a research problem for follow-up literature review and hypothesis refinement. The project starts from stream-based runtime monitoring under calibration and measurement uncertainty. Affine arithmetic tracks such uncertainty symbolically, but fresh measurement noise introduces new slack variables at each step, so the exact monitor state grows without bound. Existing zonotope reductions solve the bounded-memory problem by over-approximating the state, but they usually optimize current-set geometry rather than future monitor verdict quality. The project idea is to treat zonotope reduction as a certified abstraction-control action: an untrusted static, predictive, or learned policy chooses among certified reducers, while the reducer certificate alone preserves soundness and bounded memory.

1 Purpose of This Document

This document is designed as input for a deep-research pass. It does not try to be a paper draft. Instead, it records the project’s motivation, technical contract, implementation shape, current empirical status, and the literature questions that need better grounding before the work is written for TACAS or a similar venue.

The companion advisor memo generated under `results/tacas-main/advisor-report/` contains the numerical tables for the latest evaluation. This brief focuses on why the problem is interesting, how the current prototype maps to prior work, and where deeper research should look for novelty, threats, and stronger comparisons.

2 Problem Motivation

Stream-based runtime monitors transform input streams into derived output streams and trigger verdicts. In safety-critical settings, sensor measurements are rarely exact: some errors are persistent calibration biases, while others are independent measurement noise. The paper “Cutting Corners on Uncertainty” shows that affine arithmetic can track these uncertainties in monitor state, and that the affine coefficients form a zonotope

$$Z = c + G[-1, 1]^m.$$

Here, the center c stores nominal stream values and each generator column in G stores one symbolic uncertainty direction. Persistent calibration error can reuse a symbolic variable, but independent measurement noise creates a fresh generator at each step. Therefore, an exact monitor state grows with trace length.

The bounded-memory problem is not optional. Runtime monitors are intended to run continuously beside a system. A trace-length-dependent memory footprint is at odds with the usual runtime-assurance expectation that monitor resource requirements are statically bounded. The baseline solution is to reduce or unify generators online, producing a zonotope that encloses the exact affine state while keeping at most K generators.

3 Core Idea

The project’s central move is to stop treating reduction as a fixed local operator and instead treat it as an abstraction-control decision. At each over-budget state, a policy chooses an action from a finite set of certified reducers:

$$a_t = \pi(H_t, Z_t, \hat{x}_{t:t+h}), \quad Z'_t = \rho_{a_t, K}(Z_t).$$

The policy can use the current zonotope, trigger geometry, metadata about calibration and measurement generators, and a short predicted future trace. In the predictive variants, it evaluates candidate first reductions by rolling the monitor forward over predicted inputs and scoring future trigger imprecision. Only the first action is applied, then the monitor replans at the next real overflow.

This is model-predictive control in structure, but the control input is not a physical actuator. It is a compression choice. The objective is not tracking a desired plant trajectory; it is preserving future monitor precision under a fixed generator budget.

4 Trusted Boundary

The trusted boundary is deliberately small:

$$Z \subseteq \rho_{a, K}(Z), \quad \text{gen}(\rho_{a, K}(Z)) \leq K.$$

If every candidate reducer returns a certificate for this contract, then any policy selecting among those candidates preserves soundness. The selector can be heuristic, approximate, learned, adversarially bad for precision, or based on an inaccurate predictor; it still cannot make the monitor unsound unless it is allowed to bypass the certified reducer contract.

This is the core theorem shape to preserve in the eventual paper:

Assume the unreduced symbolic monitor transition soundly encloses all concrete states represented by its input zonotope. If every applied reduction returns a sound bounded over-approximation, then the reduced execution over-approximates the unreduced execution independently of the policy used to choose reducers.

The implementation also tracks generator metadata. Calibration, measurement, synthetic, and unknown generator columns are tagged. A monitor can declare generator requirements; `ProtectedReducer` copies required generators exactly into the reduced state and only compresses the residual. If the required columns alone exceed the budget, the reducer fails rather than silently dropping semantically required uncertainty.

5 Implementation Map

The Python prototype is organized around four boundaries:

- **Core zonotopes and certificates.** `pzr.core` stores zonotope values, generator metadata, and reduction certificates.
- **Reducers.** `pzr.reduction` implements static and monitor-aware reducers including box, Girard, Combastel, MethA, Scott, PCA, adaptive, scored-keep, identity, target-budget, and protected wrappers.
- **Monitor adapters.** `pzr.monitoring` exposes a black-box boundary: the controller can step, clone, inspect trigger metadata, and replace the zonotope component, but it does not depend on monitor equations.

- **Control policies.** `pzr.control` contains static, one-action MPC, sequence MPC, and rollout MPC selectors.

The current benchmark scenarios are:

- **robot:** the harder omnidirectional robot monitor, tracking filtered acceleration, velocity, distance increment, and x/y position.
- **robot_simple:** the two-axis motivating robot with persistent x/y calibration generators and fresh measurement generators.
- **thermostat:** a non-robot monitor with temperature, filtered temperature, heating/cooling effort, comfort deviation, and safety triggers.

There are two naming layers to keep straight. Code-level method IDs include `mpc_focused_fixed_girard` and `mpc_wide_fixed_girard`. Latest TACAS-main aggregate artifacts expose the corresponding report-facing names `mpc_rollout_girard` and `mpc_rollout_wide`. In this document, MPC-G refers to the focused fixed-Girard rollout and MPC-W to the wide first-action rollout.

Table 1: Current method roles.

Method family	Role	Soundness dependency
Static reducers	Baseline local compression	Reducer certificate
MPC-G	Focused predictive first-action search	Protected certified reducer
MPC-W	Broad first-action ablation	Protected certified reducer
Learned distilled	Cheap selector approximation	Certified fallback/reducer set
Reference	Unbounded precision reference	Not a bounded competitor

6 Background Work to Investigate

6.1 Stream-Based Runtime Monitoring

The stream-monitoring lineage is the semantic home for this project. Lola and RTLola model monitors as stream transformations, with trigger streams used for verdicts. Recent RTLola work emphasizes automatic monitor construction, static memory bounds, runtime safety, and use in cyber-physical settings. The uncertainty paper extends this setting by treating sensor values as uncertain quantities rather than exact inputs.

Deep-research questions:

- How have stream-monitoring languages formalized bounded-memory requirements, and can our generator budget be presented as a compatible resource bound?
- Are there prior runtime-monitoring papers that optimize monitor abstractions dynamically, rather than statically compiling a fixed abstraction?
- What terminology will be legible to TACAS reviewers: approximation control, abstraction control, online abstraction refinement, or monitor-aware order reduction?

6.2 Zonotope Order Reduction

Classical zonotope order reduction aims to enclose a high-order zonotope by a lower-order one with acceptable over-approximation. Girard-style, Combastel-style, PCA-based, and Scott-style reductions come from this literature. Yang and Scott compare reduction techniques by computational cost and overestimation error; Kopetzki, Schuermann, and Althoff survey known methods and introduce PCA, clustering, and optimization-based variants.

The gap for this project is semantic and temporal. These methods generally optimize current set geometry. A runtime monitor cares about future trigger verdicts, persistent calibration variables, and the way old approximation error flows through future stream equations.

Deep-research questions:

- Which order-reduction metrics in the control literature map cleanly to our interval-hull MSE, trigger width, and false-alarm metrics?
- Is there prior work on task-aware or property-aware zonotope reduction, especially where the downstream task is verification or monitoring?
- Can protected generator preservation be related to constrained zonotopes, structured uncertainty, or parameter-varying set-membership estimation?

6.3 Set-Membership Estimation and Robust Control

Zonotopes are common in set-membership estimation, reachability analysis, fault diagnosis, and robust control because they are compact and closed under linear maps and Minkowski sums. This literature gives the mathematical vocabulary for guaranteed enclosures and bounded-complexity propagation.

The connection to this project is that monitor state behaves like a symbolic state estimator over derived quantities. However, our reduction objective is not plant-state estimation error in the usual sense. It is preservation of monitor verdict precision under stream equations and trigger predicates.

6.4 MPC and Robust/Tubed Prediction

Predictive Zonotope Reduction borrows MPC’s receding-horizon pattern: optimize over a finite predicted horizon, apply the first action, then replan. Robust MPC and tube MPC are natural analogies for future work if prediction uncertainty becomes important. At present, the latest online-versus-oracle artifacts do not show a large consistent oracle advantage, so robust prediction is better framed as a future research direction rather than a current empirical claim.

Deep-research questions:

- Is there existing theory for MPC over abstractions or compression operators rather than plant inputs?
- Can rollout MPC with a fixed tail policy be described using standard approximate dynamic programming or rollout-control terminology?
- Would a robust predictor improve selector stability, or is the main bottleneck currently the reducer action set and scoring objective?

7 Current Empirical Status

The latest artifact inspected is `results/tacas-main`. Its manifest records a paper-profile run with length 300, budget 8, horizon 6, 50 seeds, both online and oracle predictor modes, and method set `paper_plus_wide`. It covers `robot`, `robot_simple`, and `thermostat`. The aggregate diagnostics are clean: zero budget violations, zero unsound certificates, zero reduction failures, and zero explicit no-op selections.

The most defensible empirical story is:

- On the hard robot monitor, MPC-G improves precision over strong static baselines at the same budget. In online mode at budget 8, mean trigger width is 225.1 for MPC-G versus 230.1 for Girard; mean interval-hull MSE is 268.4 for MPC-G versus 352.6 for Girard.
- In the robot budget sweep, MPC-G beats Girard from budgets 8 through 20. This is the cleanest current figure-level evidence for predictive reduction.
- MPC-W is not the headline method. It searches a broader first-action space and evaluates many more sequences, but on the hard robot it is less precise than MPC-G and Girard.
- The learned policy reaches about 90.6% validation accuracy and 98.8% top-3 accuracy in the latest artifact, but the training labels are heavily Girard-dominated. It should be framed as promising infrastructure, not as the main scientific result.
- Robot-simple and thermostat are useful sanity checks, but several certified reducers saturate them. They should not be used to claim broad dominance.

8 Likely Contribution Claim

The safest paper claim is:

Bounded-memory monitoring under sensor uncertainty benefits from policy-guided certified zonotope reduction. The policy can be predictive or learned, while soundness remains policy-independent because only certified reducers may mutate the monitor state. A focused receding-horizon selector improves precision on a hard robot monitor compared with strong static order-reduction baselines.

This claim avoids overcommitting on three weak points in the current evidence: MPC-W is not dominant, learned selection is not yet a strong independent contribution, and oracle prediction is not currently essential.

9 Deep-Research Agenda

1. **Positioning.** Find the closest prior work on task-aware set approximation, monitor-aware abstraction, online abstraction control, and property-preserving order reduction.
2. **Formalization.** Express the policy-independent soundness theorem using a clean monitor transition relation, a reducer action contract, and a bounded-memory invariant.
3. **Action-set design.** Explain why the focused rollout action set works better than broad first-action search in the current artifacts. Search for theory or empirical precedent about action-space restriction in rollout control.

4. **Learning.** Decide whether the learned selector is an evaluation side story or a contribution. A stronger version likely needs balanced labels, focused-rollout labels, or a learned scorer rather than a direct classifier.
5. **Prediction.** Determine whether robust or tube-aware prediction is scientifically worth adding. The current oracle gap suggests caution.
6. **Benchmarks.** Add or tune a non-robot benchmark where static reducers do not saturate, so the evaluation is not robot-specific.

10 Search Terms for Deep Research

- stream-based runtime monitoring uncertainty affine arithmetic zonotope
- RTLola bounded memory runtime monitoring sensor uncertainty
- zonotope order reduction property aware task aware
- generator reduction monitor precision false alarms
- set-membership estimation zonotope reduction structured uncertainty
- model predictive control abstraction refinement
- rollout policy fixed tail action set approximate dynamic programming
- robust MPC tube prediction set-valued future uncertainty
- runtime verification quantitative uncertainty monitor abstraction

11 Claim Boundaries

Use the following boundaries when turning this into paper prose:

- Do claim policy-independent soundness under certified reducers.
- Do claim focused rollout improves hard-robot precision in current artifacts.
- Do not claim MPC-W is always better; it is a negative/diagnostic ablation.
- Do not claim learned selection is the main empirical advance yet.
- Do not claim oracle prediction is necessary; current evidence is mixed and small.
- Do not treat the unbounded reference as a bounded-memory competitor.

12 Selected Source Anchors

References

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